The functions of vertebral column are given below.

- It protects the spinal cord.
- It supports the head.
- It serves as an attachment for the ribs.
- It provides support and place of attachment for the pectoral and pelvic girdle.
- It provides movement for the human skeleton.
- It helps in walking and standing erect with correct posture.

c. Sternum or Rib cage

Rib cage occupies the chest region. It is a cone-shaped structure made up of twelve pairs of ribs. Ribs are attached to vertebrae at the back which curve around to form a cage. Ten pairs of ribs are attached to the breast bone at the front. Two pairs of lower ribs are free at front. These are called as free-floating ribs. Rib cage is set up in such a way that it can contract and expand during the process of breathing. Rib cage protects the underlying lungs, heart and some part of liver.

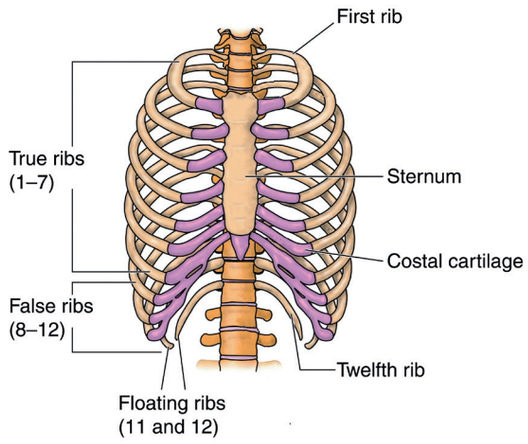


Figure 19.11 Rib cage in human



Humans and giraffes have the same number of bones in the necks, but the vertebrae in a giraffe's neck are much, much larger.

II. Appendicular skeleton

The **appendicular skeleton** contains the bones in the *appendages* of the body, as well as the structures that connect the appendages to the axial skeleton. Specifically, the appendicular skeleton comprises the shoulder girdle; the arm, wrist, and hand bones; the pelvic girdle; and the leg, ankle, and foot bones.

a. Shoulder bones or Pectoral bones

Shoulder bones are formed by collar bone at the front and the shoulder blade at the back. The collar bone is supported by breast bone at one end and the shoulder blade at the other end. The shoulder blade has a socket like cavity into which fixes the ball of the upper arm bone. This forms a ball and socket joint.

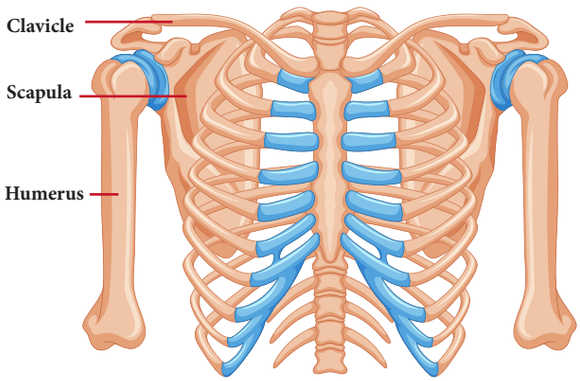


Figure 19.12 Pectoral girdles

b. Pelvic bone

Pelvic bone is also called as pelvic girdle. It is made up of strong bones to balance the entire weight of the body. Pelvic girdle is formed by five fused vertebrae at the back and form a cavity in the centre while reaching the front

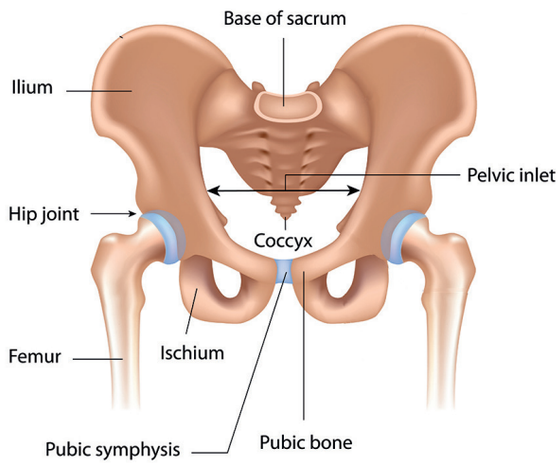


Figure 19.13 Pelvic bones

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